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Palaeomagnetic evidence for unification of the North and West Australian cratons by *ca.* 1.7 Ga: new results from the Kimberley Basin of northwestern Australia

Z. X. Li

Tectonics Special Research Centre, Department of Geology and Geophysics, the University of Western Australia, Nedlands, WA 6907, *Australia.*
E-mail: zli@tsrc.uwa.edu.au

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SUMMARY

A palaeomagnetic study of the Elgee Formation red siltstones and shales in the Palaeoproterozoic Kimberley Basin of northwestern Australia has been carried out. All seven sampling sites revealed an extremely stable magnetic remanence carried by haematite. The age of the formation is confined by precise SHRIMP U–Pb ages of early diagenetic xenotime from rocks both above and below it to be $1704 + 7/-14$ Ma, but this may represent a minimum age. The youngest detrital zircon grains in the underlying formation provide a maximum age of 1786 ± 14 Ma for the formation. The extreme stability of the remanence, the dissimilarity of the remanent direction from expected younger palaeomagnetic directions, and the lack of regional overprint in the 1790 ± 4 Ma Hart Dolerite just north of the study region support a primary origin for the remanence. A marginally positive fold test also supports a primary origin. The mean direction of $D = 92.2^\circ$, $I = 14.9^\circ$, $\alpha_{95} = 6.4^\circ$ gives a palaeopole at 4.4°S, 210.0°E with $dp = 3.3^\circ$, $dm = 6.5^\circ$. This pole, a previously reported palaeopole from the Hart Dolerite and *ca.* 1700 Ma overprint poles from the Pilbara Craton all agree with palaeopoles of similar ages from the McArthur Basin of northern Australia. Palaeomagnetic results thus suggest that the North and West Australian cratons were possibly joined together by approximately 1.7 Ga.

Key words: Australia, Elgee Formation, Kimberley, palaeomagnetism, Precambrian.

INTRODUCTION

Continental palaeomagnetism has been a major source of information for reconstructing Neoproterozoic and Phanerozoic continental configurations. However, attempts to extend such work to earlier geological times have led to vastly different forms of palaeocontinental configurations, ranging from those similar to present-day relative positions (Embleton & Schmidt 1979) to various forms of supercontinent (e.g. Piper 1982; Idnurm & Giddings 1995). Such discrepancies are largely due to the sparsity of available data, and the errors related to either the positions or the ages of the Precambrian palaeopoles or both. Because reliable palaeomagnetic poles of any given age from all cratons/continents concerned are often lacking, the common approach has been to see if a single apparent polar wander path (APWP) can be drawn for data from all the cratons/continents without violating their age sequence. However, with new dating technologies such as SHRIMP (sensitive high-resolution ion microprobe), and the gradual accumulation of high-quality palaeomagnetic results, more robust palaeomagnetic examination of Precambrian palaeoreconstructions is becoming possible. This paper reports a high-quality $1704 + 7/-14$ Ma palaeomagnetic pole from the Kimberley Craton of northwestern Australia and examines the relative position of this craton with respect to other parts of Australia at that time.

REGIONAL GEOLOGY AND PREVIOUS WORK

The Australian continent consists of numerous Archaean to Proterozoic cratons surrounded by Proterozoic orogenic belts (Fig. 1). Some believe that the relative positions of these cratons have changed little since the Proterozoic (e.g. Etheridge *et al.* 1987; Idnurm & Giddings 1988), whereas others regard many of the orogenic belts as the consequence of the closure of oceans as young as 1300–1100 Myr (e.g. Myers *et al.* 1996). Idnurm & Giddings (1988) and Idnurm *et al.* (1995) were able to draw a single APWP for Precambrian Australia, thus favouring a coherent Australian craton since Archaean time.

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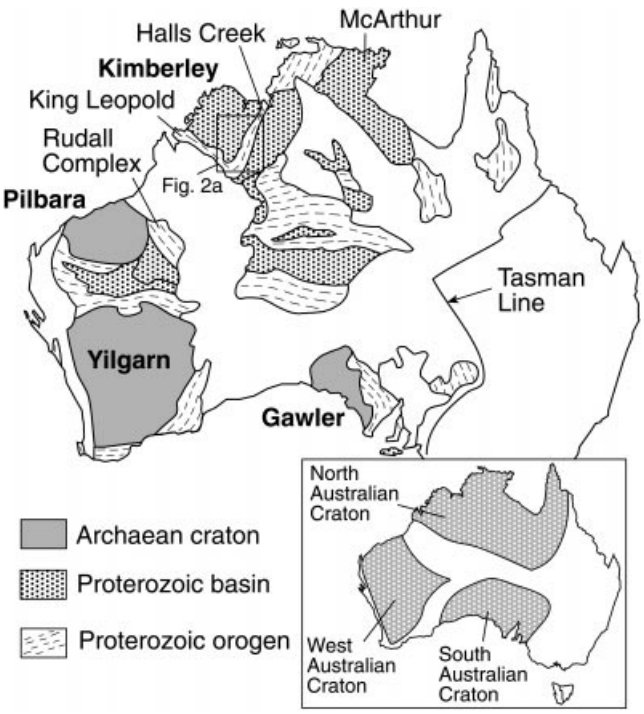


Figure 1. Exposed Australian Precambrian cratons and orogenic belts, with insert showing the locations of the North, West and South Australian cratons (after Myers *et al.* 1996).

The Kimberley Craton in northwestern Australia has been inferred to have an Archaean basement (e.g. Etheridge *et al.* 1987; Graham *et al.* 1999), although recent sedimentary provenance analysis of Palaeoproterozoic metasediments, probably derived from the Kimberley Craton, showed no evidence of such a basement (Tyler 1999). The Craton is bounded to the southwest by the King Leopold Orogen and to the southeast by the Halls Creek Orogen (Fig. 1). A number of orogenic events have been recognized in one or both orogenic belts: the Hooper Orogeny (1865–1850 Ma), the Halls Creek Orogeny (1830–1800 Ma), the Yampi Orogeny (1300–1000 Ma), the King Leopold Orogeny (approximately 560 Ma) and the Alice Springs Orogeny (400–300 Ma) (Tyler & Griffin 1994; Tyler *et al.* 1995; Tyler & Page 1996; Myers *et al.* 1996). The study region is at the southeastern corner of the craton, on the southern limb of the O'Donnell Syncline (Figs 1 and 2a), where the Kimberley Group unconformably overlies the $\geq$1800 Ma crystalline basement (the Lamboo Complex) and the Speewah Group (Fig. 2b).

The Palaeoproterozoic sedimentary/volcanic successions in the Kimberley Basin have been well dated (Fig. 2b). Felsic volcanics interbedded with sediments in the Valentine Siltstone of the Speewah Group give a SHRIMP U–Pb zircon age of 1834 ± 3 Ma (Page & Sun 1994). The Hart Dolerite, which intruded the lower Kimberley Group (McNaughton *et al.* 1999; I. Tyler, personal communication, 1999), has been dated by the SHRIMP U–Pb zircon method at 1790 ± 4 Myr (R.W. Page, AGSO OZCHRON database). McNaughton *et al.* (1999) recently reported 1704 ± 7 and 1704 ± 14 Ma SHRIMP U–Pb ages for early diagenetic xenotime from the Warton and Pentecost sandstones, respectively, which thus constrain the age of the Elgee Siltstone to $1704 + 7/-14$ Ma. However, regional correlation suggests that this could represent a minimum age (I. Tyler, personal communication, 1999). A maximum age of 1786 ± 14 Ma is provided by the age of the youngest detrital zircon grains in the underlying Warton Sandstone (McNaughton *et al.* 1999) (Fig. 2b).

McElhinny & Evans (1976) reported palaeomagnetism from the Hart Dolerite just north of the study region. They revealed an east-southeasterly subhorizontal remanence direction that passed a positive baked contact test. The remanence was interpreted as being of primary origin, and the corresponding palaeopole is at 29°N, 46°E with $A_{95} = 24°$. Other published Precambrian palaeopoles for the Kimberley Craton include the WTC pole from the cap dolomite of the Neoproterozoic Walsh Tillite about 150 km northwest of the study region, which passed a fold test (Li 2000), and a pole from the Early

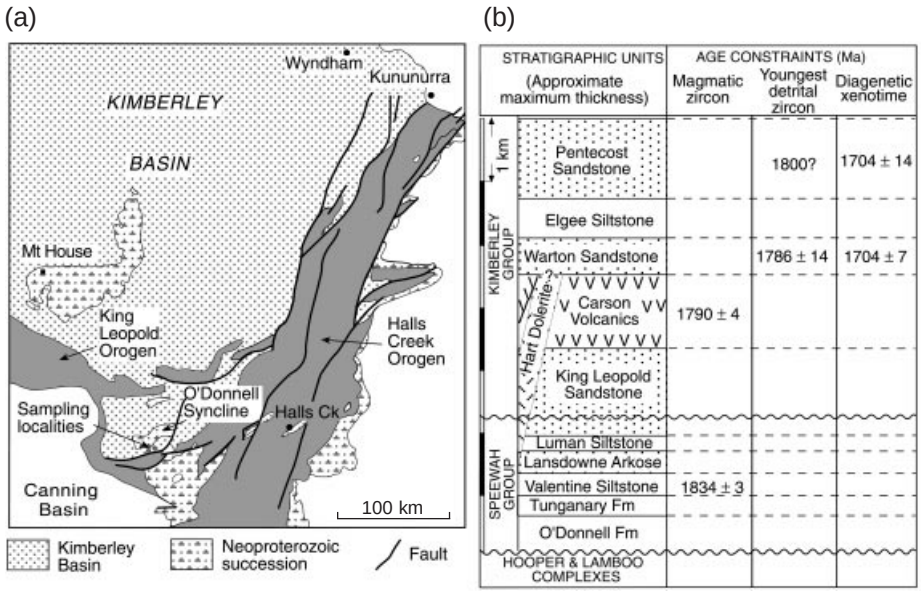


Figure 2. (a) Sampling localities and (b) stratigraphic column of the Kimberley Basin and available age constraints (after McNaughton *et al.* 1999).

Cambrian Antrim Plateau Volcanics in northeastern Kimberley and adjacent regions to the east (McElhinny & Luck 1970). No pervasive regional overprint was reported in either study.

The target of this study, the Elgee Formation, consists of purple siltstone and silty shale with minor quartz sandstone. In the sampling region, the formation is gently tilted with dip angles ranging between 8° and 19°. Although cleavages are visible in some localities, they were avoided during palaeomagnetic sampling. A total of 55 oriented samples were collected from seven outcrops (sites) (Fig. 2a). Both a sun compass and a magnetic compass were used for orienting the samples.

DEMAGNETIZATION RESULTS AND AGE OF THE CHARACTERISTIC REMANENCE

All samples were analysed in the palaeomagnetism laboratory of The University of Western Australia. One specimen from each sample was subjected to stepwise thermal demagnetization using two MMTD1 furnaces manufactured by Magnetic Measurements, and remanence was measured using a 2G 755-R cryogenic magnetometer. A few specimens were also subjected to stepwise alternating field (AF) demagnetization using an automated 2G600 AF demagnetizer attached to the cryogenic magnetometer, but this method proved to be ineffective in delineating remanence vectors. All demagnetization and remanence measurements were carried out within a low-field space created by three sets of 3.8 m × 3.8 m Helmholtz coils.

Almost all samples (96 per cent) had an extremely stable high-temperature remanence vector (Figs 3 and 4). Although a small low-temperature component, most probably due to recent weathering, is visible in many specimens, it is prominent enough to give a site mean direction for two of the seven sites only (Table 1). The unblocking temperature of the characteristic remanence is always above 600 °C, indicating haematite as the magnetic carrier. The low-temperature component is likely to be carried by both goethite and very fine-grained haematite.

After discarding the high-temperature component of two samples as outliers, the site mean directions of all seven sites have α_{95} equal to or below 10° (Figs 4 and 5, Table 1). An extended McFadden & Jones test (McFadden & Jones 1981; McFadden 1998) was applied to the seven sites, which were divided into two populations of three and four according to their bedding attitudes. However, the test proved to be inconclusive with the test statistics f equal to 0.767 *in situ* (the probability of exceeding 0.767 is 49 per cent) and 0.261 after 100 per cent bedding correction (the probability of exceeding 0.261 is 78 per cent). A McFadden (1990) correlation fold test was also applied to these site mean directions. Test definition 1 was again inconclusive, possibly because the bedding tilt directions were not divergent enough. However, a definition 2 fold test proved to be positive at the 95 per cent confidence level. As listed in Table 1, the test statistics SCOS (95 per cent) are 3.086, whereas the *in situ* SCOS2 value (3.131) is above

Table 1. List of site mean directions and palaeopoles from the Elgee Formation (location 18.4°S, 126.6°E).

Primary remanence

Site	*n*/no	*In situ*		Bedding dip	Bedding-corrected		Common	
		Dec (°)	Inc (°)	azimuth/dip (°)	Dec (°)	Inc (°)	α_{95} (°)	*k*
KE01	8/10	94.9	10.0	009/19	91.9	8.1	10.4	29.1
KE02	13/14	91.6	8.8	009/19	89.2	5.9	8.1	26.9
KE03	8/8	99.8	13.8	322/13	96.8	23.2	7.4	57.2
KE04	9/10	92.7	−0.1	338/19	91.5	7.7	8.3	39.2
KE05	5/5	96.1	6.2	261/15	97.1	20.7	3.6	457.2
KE06	4/4	89.9	4.1	243/11	90.7	13.9	7.7	143.3
KE07	4/4	92.2	22.1	340/8	88.9	24.9	5.4	287.7
Total mean		93.9	9.3		92.2	14.9		
		$\alpha_{95} = 5.9°$			$\alpha_{95} = 6.4°$			
		$k = 105.6$			$k = 89.1$			
McFadden (1990) correlation test definition 2 results:								
$N = 7$, SCOS 95% = 3.086, SCOS 99% = 4.253								
		in situ			bedding-corrected			
		SCOS2 = 3.131			SCOS2 = 1.136			
Palaeopole position		5.1°S, 213.3°E			4.4°S, 210.0°E			
		$dp = 3.0°$, $dm = 5.9°$			$dp = 3.3°$, $dm = 6.5°$			
					(pole EG)			

Remanence of recent origin

Site	*n*/*no*	Dec (°)	Inc (°)	α_{95} (°)	*k*
KE02	6/14	2.1	−64.2	21.4	10.8
KE04	8/10	359.3	−41.8	6.8	68.1

n/*no*: number of samples showing the characteristic remanence/total number of samples; *N*: number of sites; α_{95}: half-angle of the 95 per cent confidence cone for the mean direction; Dec: declination; Inc: inclination; *k*: precision parameter of Fisher (1953) distribution for mean directions; *dp*/*dm*: semi-axes of error ellipse around the pole at a probability of 95 per cent; SCOS: critical value of McFadden (1990) correlation test statistics, where '2' stands for definition 2.

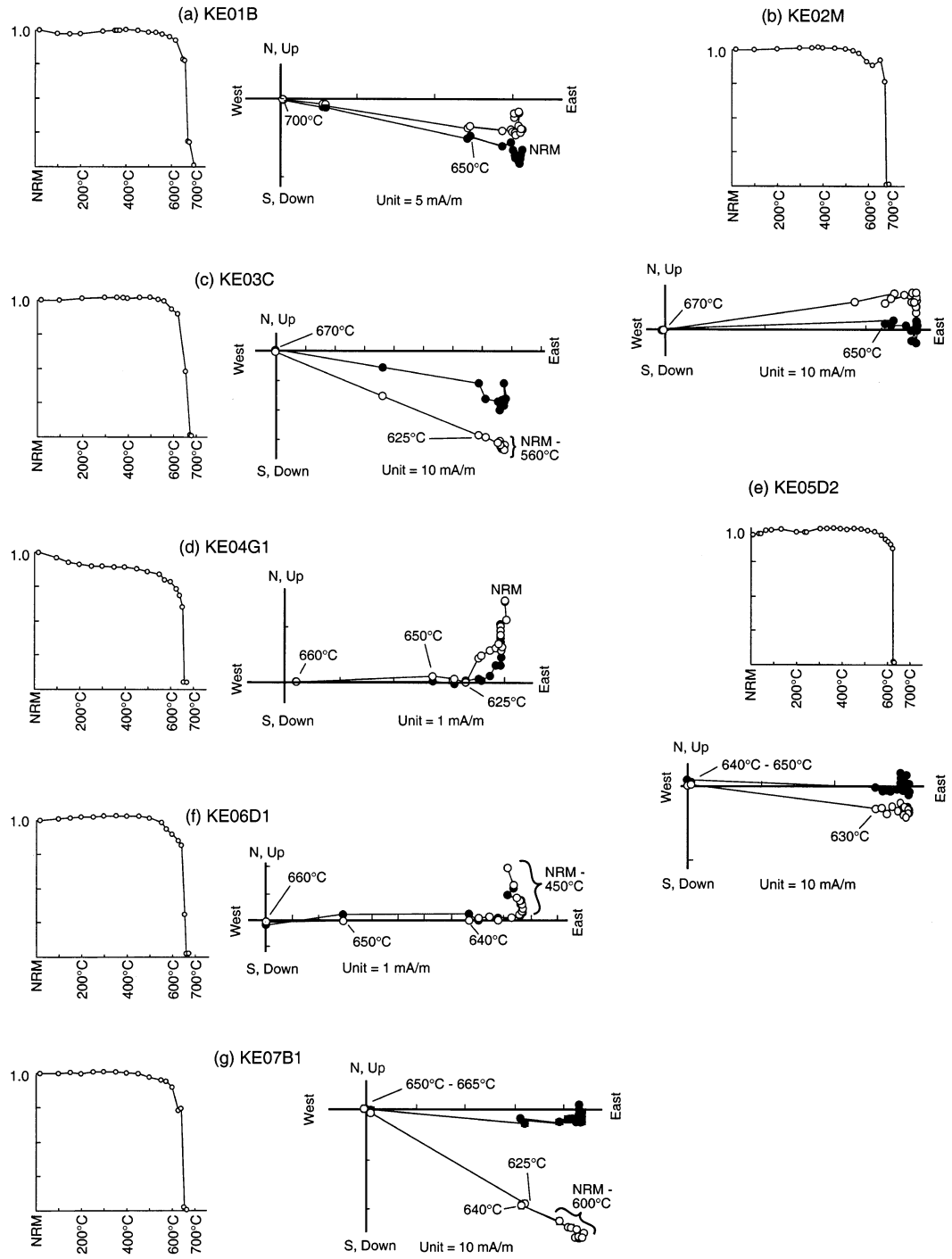


Figure 3. Representative demagnetization results of one sample from each site showing both normalized intensity decay curves of the natural remanent magnetization and orthogonal plots. Solid (open) circles in the orthogonal plots are projections on the horizontal (vertical) plane.

this critical value, suggesting that the remanence was probably not acquired after deformation. On the other hand, the bedding-corrected SCOS2 value (1.136) is below the critical value, suggesting that it is more likely that the remanence was acquired before any deformation.

The result of the fold test is not clear-cut, and should be treated with caution. Nonetheless, since the bedding tilt angles are low, bedding correction makes little difference to the overall mean direction (Fig. 5), and there is thus little palaeohorizontal uncertainty in the remanence directions.

The exact timing of the gentle deformation in the studied region is uncertain. The 1300–1000 Ma Yampi Orogeny and the *ca.* 560 Ma King Leopold Orogeny are the two known compressional events along the southern Kimberley Craton since the deposition of the sediments (e.g. Tyler & Griffin 1990; Myers *et al.* 1996), and both could have caused the gentle folding. Pre-folding remanence should thus be older than 560 Ma, and possibly older than 1300–1100 Ma.

Although we could not establish the primary origin of the remanence based on the fold test alone, the following points

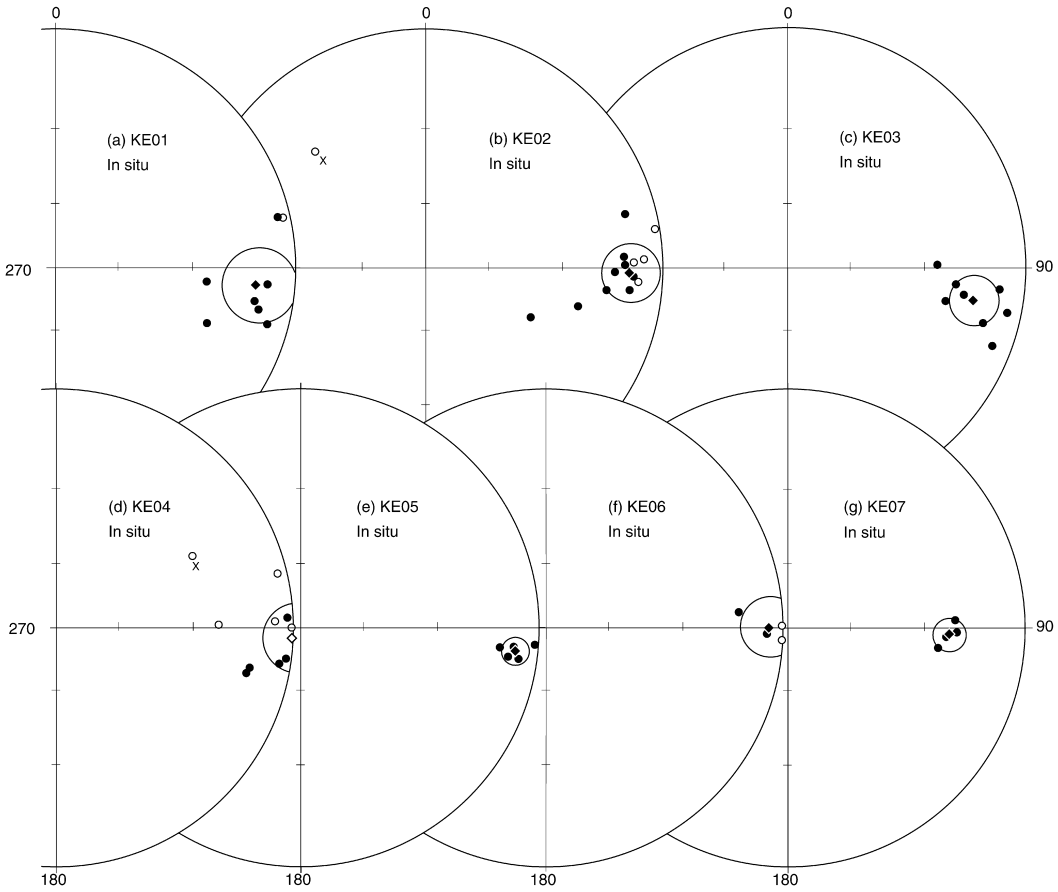


Figure 4. Equal-angle projections of remanence directions from each site. Solid (open) symbols are projections on the lower (upper) hemisphere. Each circular data point represents one sample. Diamonds and small circles represent site mean directions and their α_{95}. Samples marked with an '×' are considered outliers, and excluded from the mean direction calculation.

suggest that the remanence is of primary (or early diagenetic) origin. (1) The extreme high stability of the remanence can only be overprinted by temperatures over 600 °C or a pervasive chemical event (e.g. Embleton & Williams 1986), neither of which is evident in the study region. (2) The remanence direction is different from expected younger directions based on published younger palaeomagnetic poles from Australia. (3) No regional magnetic overprint has been reported from either the 1790 ± 4 Ma Hart Dolerite just north of the study region (McElhinny & Evans 1976) or the cap dolomite of the Neoproterozoic Walsh Tillite northwest of the study region (Li 2000). The age of the remanence is therefore regarded as close to the age of the rock formation, that is, between 1786 ± 14 and $1704 + 7/-14$ Ma.

PALAEOMAGNETIC POLE POSITIONS AND TECTONIC IMPLICATIONS

Ca. 1800 Ma amalgamation of the Kimberley Craton and the eastern North Australian Craton

The palaeomagnetic pole (pole EG) corresponding to the bedding-corrected mean direction is at 4.4°S, 210°E with $dp = 3.3°$ and $dm = 6.5°$ (Table 1). This pole and the HD pole from the Hart Dolerite (McElhinny & Evans 1976) are plotted against the recently published Palaeoproterozoic apparent polar wander path (APWP) for the McArthur Basin of Idnurm *et al.* (1995) (Fig. 6a).

As shown in Fig. 6, the 95 per cent confidence oval of the EG pole from the Elgee Formation overlaps with the A_{95} of the HR pole from the 1725 ± 2 Ma Hobblechain Rhyolite (Idnurm *et al.* 1995; Rawlings 1999). The EG pole also falls close to the WKF pole from the Wollogorang Formation (Idnurm *et al.* 1995), dated at 1730 ± 3, 1729 ± 4 and 1723 ± 4 Myr using the SHRIMP U–Pb zircon method (see summary in Fig. 3 of Rawlings 1999). The minimum age of $1704 + 7/-14$ Ma for the EG pole is slightly younger than the ages for the HR and WFK poles, but its relative position is consistent with its being a slightly younger pole (Fig. 6a).

The only other Palaeoproterozoic palaeomagnetic pole from the Kimberley Craton, the 1790 ± 4 Ma HD pole, has a large error, but the pole position nonetheless falls on the older end of the McArthur Basin APWP, a position again consistent with its age. The two poles from the Kimberley Craton may thus share the same APWP as the McArthur Basin. A slightly modified APWP based on both data sets is given in Fig. 6(b). Note that due to the poor age constraints on the PMGO and HRO poles—both overprint poles from the McArthur Basin (Idnurm *et al.* 1995)—slightly different shapes for the 1800–1700 Ma APWP loop are possible.

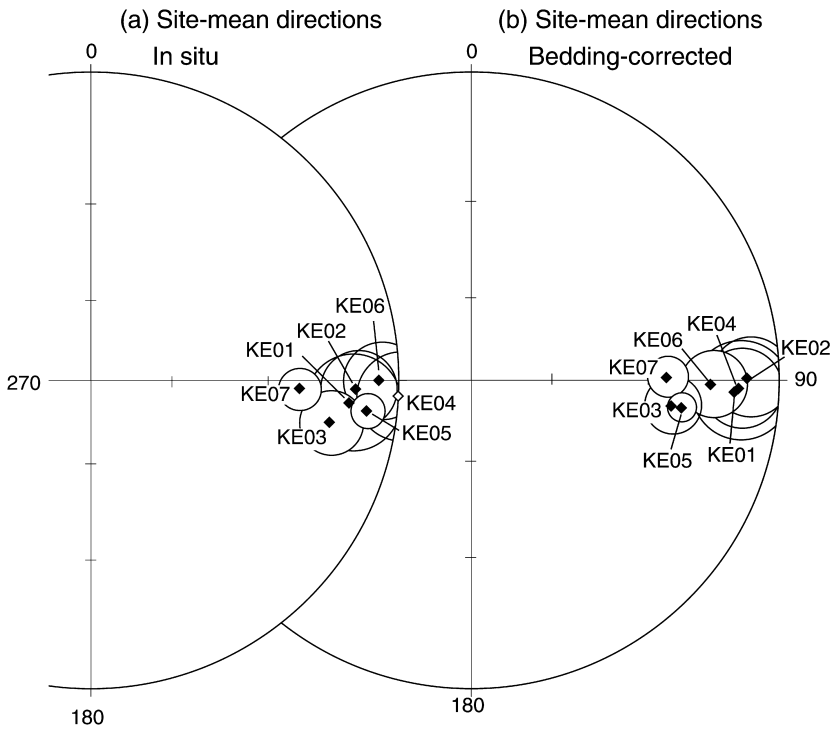


Figure 5. Equal-angle projections of site mean directions and their α_{95} (a) *in situ* and (b) bedding-corrected. Solid (open) symbols are projections on the lower (upper) hemisphere.

The coherent palaeomagnetic results across the Halls Creek Orogeny support the geological argument that the Kimberley Craton had joined the North Australian Craton by ~1800 Ma (e.g. Tyler & Page 1996; Myers *et al.* 1996) (Fig. 1, insert). So far, there are no older palaeomagnetic data available from the two cratons that would allow an examination of their earlier relationship.

Had the West Australian Craton joined the North Australian Craton by ~1700 Ma?

Also plotted in Fig. 6(b) are the syn- to post-folding magnetic overprint poles from the Hamersley Province of southern Pilbara, where JO is the synfolding overprint pole from the Mount Jope Volcanics (Schmidt & Embleton 1985), PBA the *in situ* pole position from the iron ores at Paraburdoo (Schmidt & Clark 1994), TPA the *in situ* pole position from the iron ores at Tom Price, and HP2 the overprint pole based on data mostly from carbonate rocks in the Hamersley Province (Li *et al.* 1993; Li *et al.*, unpublished data). These overprint poles are likely to be related to the last major orogeny that had significant impact on the Hamersley Province, the Capricorn Orogeny, which peaked at around 1840–1790 Ma, with deformation lasting until *ca.* 1700 Ma (Myers *et al.* 1996; Tyler 1999). Since the overprint mostly appears to be of post-folding origin, its age is likely to be around 1790–1700 Ma or slightly younger.

Fig. 6(b) shows that the 95 per cent confidence ovals of A_{95} of the poles from the Pilbara all fall within that of the 1790 Ma HD pole from the Kimberley Craton. However, they also overlap with the MAL pole from the McArthur Basin, which has an age of between 1713 ± 7 and 1640 ± 3 Ma (Idnurm *et al.* 1995; Rawlings 1999). The poor age constraints for the Pilbara overprint poles make both ages possible, but in any case they do not require any ocean between the North and West Australian cratons during the Mesoproterozoic, as postulated by Myers *et al.* (1996) and Tyler *et al.* (1998). However, the possibility of such an ocean cannot be totally ruled out because the well-grouped Pilbara poles give no constraints on the relative palaeolongitudes of the two cratons.

Smithies & Bagas (1997) reported evidence from the Rudall Complex (Fig. 1) of a continental collision between the Western Australian Craton and an unknown continent to the north-east during the Yapungku Orogeny (2000–1760 Ma), which caused high-pressure metamorphism between 1790 and 1760 Ma. Magmatic arcs and orogenic events of similar age exist along the southern margin of the North Australian Craton (Myers *et al.* 1996; Smithies & Bagas 1997), and can be interpreted as evidence that the Northern Australian Craton was the continent that collided with the West Australian Craton. Furthermore, there is no geological evidence for any ocean between the two cratons after *ca.* 1700 Ma other than the ~1300 Ma granitic intrusions and some tectonic reactivation in the Rudall Complex (e.g. Myers *et al.* 1996; Smithies & Bagas 1997). It is therefore geologically possible that the West and North Australian cratons joined together by *ca.* 1700 Ma. There is as yet no 1800–1700 Ma pole from the South Australia Craton that would allow a similar examination of its relative position.

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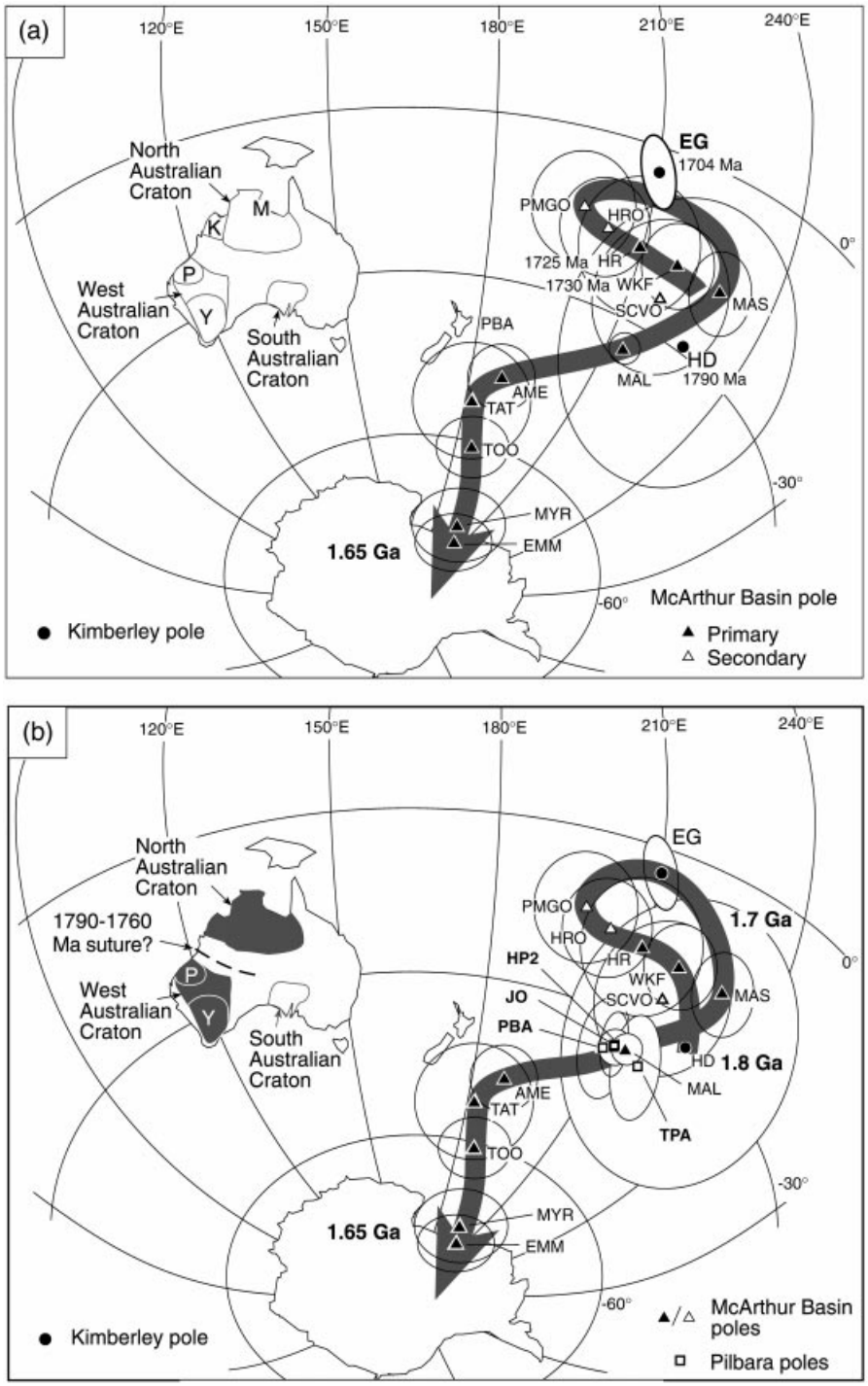


Figure 6. (a) Palaeopoles from the Kimberley Craton (K) plotted against the Palaeoproterozoic palaeopoles and apparent polar wander path of the McArthur Basin (M). EG: Elgee Formation pole (this study); HD: Hart Dolerite pole (McElhinny & Evans 1976). For details of the McArthur Basin poles (shown as triangles) see text discussion and Table 2 of Idnurm *et al.* (1995). Updated ages for the McArthur Basin poles are from Fig. 3 of Rawlings (1999). P: Pilbara Craton; Y: Yilgarn Craton. (b) Late to post-Capricorn Orogeny overprint poles from southern Pilbara (squares) plotted against the North Australian cratons palaeopoles and apparent polar wander path. JO: synfolding overprint pole from the Mount Jope Volcanics (Schmidt & Embleton 1985); PBA: *in situ* pole position from iron ores at Paraburdoo (Schmidt & Clark 1994); TPA: *in situ* pole position from iron ores at Tom Price; HP2: late, mostly post-folding, overprint pole from the Hamersley Province (Li *et al.* 1993; Li *et al.*, unpublished data).

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